

Complete resolution of equation (4): a sharp jump at $q = 1$

Autonomous proof candidate for specialist verification

9 August 2026

Abstract

Let $1 < p < 2$ and $1 \leq q \leq 2$. We resolve equation (4) in Blasco–Botelho–Pellegrino–Rueda: every continuous bilinear form on $\ell_p \times \ell_q$ is absolutely $(p; 2, 1)$ -summing if and only if $q = 1$. At $q = 1$ we prove the stronger $(1; 2, 1)$ conclusion using Grothendieck factorization, the little Grothendieck theorem, and trace ideals. For every $1 < q \leq 2$ we give an explicit compact-operator counterexample built from uniformly complemented Rademacher Euclidean blocks. The literature search supporting novelty was bounded, so the result should be treated as a full candidate pending specialist review.

1 Source question and answer

For a Banach space E and a finite family $(z_j)_{j=1}^m \subset E$, write

$$\|(z_j)_{j=1}^m\|_{\ell_t^w(E)} = \sup_{\varphi \in B_{E^*}} \left(\sum_{j=1}^m |\varphi(z_j)|^t \right)^{1/t}.$$

A continuous bilinear form $A : E \times F \rightarrow \mathbb{K}$ is absolutely $(u; r, s)$ -summing if a constant C exists such that

$$\left(\sum_{j=1}^m |A(x_j, y_j)|^u \right)^{1/u} \leq C \| (x_j) \|_{\ell_r^w(E)} \| (y_j) \|_{\ell_s^w(F)} \quad (1)$$

for all finite families. The least such constant is denoted $\pi_{(u;r,s)}(A)$.

On page 9 of the source paper [1], after proving nearby coincidence results, the authors ask whether, for $1 < p < 2$ and $1 \leq q \leq 2$,

$$\mathcal{L}(\ell_p, \ell_q; \mathbb{K}) = \Pi_{(p;2,1)}(\ell_p, \ell_q; \mathbb{K}). \quad (2)$$

The source text is reproduced below.

for $1 \leq q \leq 2$. Whereas, from the Inclusion Theorem and the Defant-Voigt Theorem, one also has, for any $1 \leq q < \infty$,

$$\mathcal{L}(\ell_2, \ell_q; \mathbb{K}) = \Pi_{(2;2,1)}(\ell_2, \ell_q; \mathbb{K}).$$

So, by interpolation one may expect that for $1 < p < 2$ and $1 \leq q \leq 2$,

$$\mathcal{L}(\ell_p, \ell_q; \mathbb{K}) = \Pi_{(p;2,1)}(\ell_p, \ell_q; \mathbb{K}). \quad (4)$$

We do not know if (4) holds, but we shall prove now that it does not hold for $q > 2$ (cf. Proposition 4.3) and, moreover, next we will see that we can get quite close to (4) via interpolation (cf. Theorem 4.5).

Our answer is a sharp endpoint classification.

Theorem 1 (Full resolution of (2)). *Let $1 < p < 2$ and $1 \leq q \leq 2$. Then*

$$\mathcal{L}(\ell_p, \ell_q; \mathbb{K}) = \Pi_{(p;2,1)}(\ell_p, \ell_q; \mathbb{K}) \iff q = 1.$$

More precisely:

1. *every $A \in \mathcal{L}(\ell_p, \ell_1; \mathbb{K})$ is absolutely $(1; 2, 1)$ -summing;*
2. *for every $1 < q \leq 2$, there is a compact operator $T : \ell_q \rightarrow \ell_{p'}$ such that $A_T(x, y) = \langle Ty, x \rangle$ is not absolutely $(p; 2, 1)$ -summing.*

2 Proof intuition

At $q = 1$, a weakly ℓ_1 family is encoded by an operator $R : \ell_\infty^m \rightarrow \ell_1$. Grothendieck's factorization theorem factors R through a Hilbert space. One resulting column operator is Hilbert–Schmidt by averaging over signs. The little Grothendieck theorem makes the other constructed operator Hilbert–Schmidt. Their product is trace class, and its diagonal is exactly $(A(x_j, y_j))_j$. This gives an ℓ_1 estimate, stronger than the required ℓ_p estimate.

When $q > 1$, both ℓ_q and $\ell_{p'}$ contain uniformly complemented Euclidean spaces of every finite dimension. We use explicit Rademacher embeddings and projections. A block-diagonal operator which is a slowly decaying scalar multiple of the identity on these Euclidean blocks is compact, yet carefully chosen diagonal test sequences force the quotient in (1) to diverge.

3 The positive endpoint $q = 1$

We use two standard forms of Grothendieck's theorem; see, for example, Pisier's survey [2]. First, every operator $R : \ell_\infty^m \rightarrow \ell_1$ admits a Hilbert-space factorization

$$\ell_\infty^m \xrightarrow{J} H \xrightarrow{V} \ell_1, \quad R = VJ, \quad \|V\| \|J\| \leq K_G \|R\|. \quad (3)$$

Second, the little Grothendieck theorem says that every $S : \ell_\infty \rightarrow H$ is 2-summing and

$$\pi_2(S) \leq K_G \|S\|. \quad (4)$$

The constants may depend on the scalar field and their exact values are irrelevant.

Fix $A \in \mathcal{L}(\ell_p, \ell_1; \mathbb{K})$, and let $T : \ell_1 \rightarrow \ell_{p'}$ be its associated operator: $A(x, y) = \langle Ty, x \rangle$. Given finite families $(x_j)_{j=1}^m \subset \ell_p$ and $(y_j)_{j=1}^m \subset \ell_1$, define

$$X : \ell_2^m \rightarrow \ell_p, \quad Xe_j = x_j, \quad R : \ell_\infty^m \rightarrow \ell_1, \quad Ra = \sum_{j=1}^m a_j y_j.$$

Then

$$\|X\| = \|(x_j)\|_{\ell_2^w(\ell_p)}, \quad \|R\| = \|(y_j)\|_{\ell_1^w(\ell_1)}. \quad (5)$$

Choose the factorization (3), and introduce

$$U : \ell_2^m \rightarrow H, \quad Ue_j = Je_j, \quad C = X^*TV : H \rightarrow \ell_2^m.$$

A Rademacher average gives

$$\|U\|_{S_2}^2 = \sum_{j=1}^m \|Je_j\|^2 = \mathbb{E} \|J(\varepsilon_1, \dots, \varepsilon_m)\|^2 \leq \|J\|^2. \quad (6)$$

Moreover, $C^* = V^*T^*X$. By (4), the ideal property of the 2-summing norm, and the equality between 2-summing and Hilbert–Schmidt norms for operators between Hilbert spaces,

$$\|C\|_{S_2} = \|C^*\|_{S_2} \leq K_G \|V\| \|T\| \|X\|. \quad (7)$$

Thus CU is trace class. Since the diagonal map from $S_1(\ell_2^m)$ to ℓ_1^m is contractive and

$$\langle CUe_j, e_j \rangle = \langle TVJe_j, Xe_j \rangle = \langle Ty_j, x_j \rangle = A(x_j, y_j),$$

we obtain from (3), (5)–(7)

$$\begin{aligned} \sum_{j=1}^m |A(x_j, y_j)| &\leq \|CU\|_{S_1} \leq \|C\|_{S_2} \|U\|_{S_2} \\ &\leq K_G^2 \|A\| \|(x_j)\|_{\ell_2^w(\ell_p)} \|(y_j)\|_{\ell_1^w(\ell_1)}. \end{aligned} \quad (8)$$

This proves the $(1; 2, 1)$ assertion. Since the ℓ_p norm of a finite scalar family is at most its ℓ_1 norm, the desired $(p; 2, 1)$ assertion follows.

4 Uniform Rademacher Euclidean blocks

The counterexample uses the following explicit standard consequence of the Khintchine inequality [3].

Lemma 1. *For every $1 < a < \infty$ and every positive integer d , there are maps*

$$E_{a,d} : \ell_2^d \rightarrow \ell_a^{2^d}, \quad Q_{a,d} : \ell_a^{2^d} \rightarrow \ell_2^d$$

such that $Q_{a,d}E_{a,d} = I_{\ell_2^d}$ and

$$\|E_{a,d}\| \leq B_a, \quad \|Q_{a,d}\| \leq B_{a'}, \quad (9)$$

where the constants depend only on a .

Proof. Put $N = 2^d$ and index the coordinates of $\mathbb{K}^N$ by $\varepsilon \in \{-1, 1\}^d$. Define

$$(E_{a,d}u)_\varepsilon = N^{-1/a} \sum_{j=1}^d u_j \varepsilon_j, \quad (Q_{a,d}z)_j = N^{-1/a'} \sum_{\varepsilon} z_\varepsilon \varepsilon_j. \quad (10)$$

Orthogonality of the signs and $1/a + 1/a' = 1$ give $Q_{a,d}E_{a,d} = I$ exactly. The upper Khintchine inequality gives the first bound in (9). For $v \in \ell_2^d$, Hölder's inequality followed by Khintchine gives

$$|\langle Q_{a,d}z, v \rangle| \leq \|z\|_a \left(N^{-1} \sum_{\varepsilon} \left| \sum_j v_j \varepsilon_j \right|^{a'} \right)^{1/a'} \leq B_{a'} \|z\|_a \|v\|_2,$$

which proves the second bound. □

5 A compact counterexample for every $q > 1$

Fix $1 < q \leq 2$, put $s = p' > 2$, and define $r > 0$ by

$$\frac{1}{r} = \frac{1}{p} - \frac{1}{2}. \quad (11)$$

Let $d_n = 2^n$ and $N_n = 2^{d_n}$. Apply Lemma 1 in each block:

$$E_n = E_{q,d_n}, \quad Q_n = Q_{q,d_n}, \quad F_n = E_{s,d_n}, \quad B_n = Q_{s,d_n}. \quad (12)$$

Thus $Q_n E_n = B_n F_n = I_{\ell_2^{d_n}}$, and all four operator norms are uniformly bounded (with constants depending only on p and q).

Partition the coordinates so that

$$\ell_q = \left(\bigoplus_{n \geq 1} \ell_q^{N_n} \right)_q, \quad \ell_s = \left(\bigoplus_{n \geq 1} \ell_s^{N_n} \right)_s.$$

Set

$$\lambda_n = d_n^{-1/(2r)}, \quad T = \bigoplus_{n \geq 1} \lambda_n F_n Q_n : \ell_q \rightarrow \ell_s. \quad (13)$$

This is bounded. Indeed, for $z = (z_n)$ and a constant $C = C(p, q)$,

$$\|Tz\|_s \leq C \sup_n \lambda_n \left(\sum_n \|z_n\|_q^s \right)^{1/s} \leq C \sup_n \lambda_n \|z\|_q, \quad (14)$$

because $q < s$. If $T^{(N)}$ retains only the first N blocks, it has finite rank and the same estimate gives

$$\|T - T^{(N)}\| \leq C \sup_{n > N} \lambda_n \rightarrow 0. \quad (15)$$

Consequently T is compact.

Let $A_T(x, y) = \langle Ty, x \rangle$ on $\ell_p \times \ell_q$. In the n th block, for $1 \leq j \leq d_n$, set

$$y_j^{(n)} = E_n e_j \in \ell_q, \quad x_j^{(n)} = B_n^* e_j \in \ell_p. \quad (16)$$

The weak norms satisfy

$$\left\| (x_j^{(n)})_{j=1}^{d_n} \right\|_{\ell_2^w(\ell_p)} = \|B_n^*\| = \|B_n\| \leq B_p, \quad (17)$$

and, viewing the column map as $E_n : \ell_\infty^{d_n} \rightarrow \ell_q^{N_n}$,

$$\left\| (y_j^{(n)})_{j=1}^{d_n} \right\|_{\ell_1^w(\ell_q)} \leq \left\| E_n : \ell_\infty^{d_n} \rightarrow \ell_q^{N_n} \right\| \left\| I : \ell_\infty^{d_n} \rightarrow \ell_2^{d_n} \right\| \leq B_q \sqrt{d_n}. \quad (18)$$

On the other hand, (12)–(13) give, for every j ,

$$A_T(x_j^{(n)}, y_j^{(n)}) = \lambda_n \langle F_n e_j, B_n^* e_j \rangle = \lambda_n \langle B_n F_n e_j, e_j \rangle = \lambda_n. \quad (19)$$

If A_T were absolutely $(p; 2, 1)$ -summing, (17)–(19) would force

$$\pi_{(p; 2, 1)}(A_T) \geq \frac{\lambda_n d_n^{1/p}}{B_p B_q \sqrt{d_n}} = \frac{1}{B_p B_q} d_n^{1/(2r)} \rightarrow \infty, \quad (20)$$

where (11) and the choice of λ_n were used. This contradiction proves the negative assertion and completes Theorem 1.

6 Verification and novelty limitations

The finite matrix identity in Lemma 1 and the exponent in (20) were independently checked by the included script `code/verify_rademacher_blocks.py`. The computation is not used as proof. The delicate mathematical points for review are the normed form (3) of Grothendieck factorization and the application of little Grothendieck to V^* in (7).

Exact-title, equation, and exponent searches recovered the source paper and general summability literature but no later explicit resolution of (2). This was a bounded search, not an exhaustive bibliographic proof. The classification, especially the Euclidean-block obstruction, should be checked against operator-ideal literature using other terminology before any public novelty claim.

References

- [1] O. Blasco, G. Botelho, D. Pellegrino, and P. Rueda, *Absolutely summing multilinear operators on ℓ_p spaces*, Collect. Math. 67 (2016), 519–532; arXiv:1404.1322.
- [2] G. Pisier, *Grothendieck’s theorem, past and present*, Bull. Amer. Math. Soc. 49 (2012), 237–323; arXiv:1101.4195.
- [3] U. Haagerup, *The best constants in the Khintchine inequality*, Studia Math. 70 (1981), 231–283.